Question				Solution	Marks
1	(a)	(i)		Phase difference = ¼ cycle × 2π = π/2 or 1.57 rad	A1
		(ii)	1.	2.2 mm	A1
			2.	1.6 mm + 1.6 mm = 3.2 mm	A1
				1.6 mm – 1.6 mm = 0 mm	A1
	(b)	(i)		Path difference varies between nλ and (n + ½)λ (phase difference alternates between 0 and π) Series of alternating maxima and minima OR 2 waves of same amplitude and frequency travelling in opposite directions. A stationary wave is formed. Series of alternating maxima at the antinodes and minima at the nodes	M1 A1 (M1) (A1)
		(ii)		Path difference = 0 (waves meet in phase) Maximum throughout	M1 A1
				Path difference = 3.5λ (waves meet in antiphase) Minimum throughout	M1 A1

Question			Solution	Marks
2	(a)	(i)	$2mgR$	